Third (backward) differences:

$$\nabla^3 y_n = \nabla^2 y_n - \nabla^2 y_{n-1} \quad n = 1,2,3,\ldots$$

In general, k^{th} (backward) differences:

$$\nabla^k y_n = \nabla^{k-1} y_n - \nabla^{k-1} y_{n-1} \quad n = 1,2,3,\ldots$$

Backward difference table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
		∇y_1			
x_1	y_1		$\nabla^2 y_2$		
		∇y_2		$\nabla^3 y_3$	
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$
		∇y_3		$\nabla^3 y_4$	
x_3	y_3		$\nabla^2 y_4$		
		∇y_4			
x_4	y_4				

Backward differences can also be defined as follows.

$$\nabla f(x) = f(x) - f(x-h)$$

First differences:

$$\nabla f(x+h) = f(x+h) - f(x)$$

$$\nabla f(x+2h) = f(x+2h) - f(x+h),$$

h is the interval of spacing.

Second differences:

$$\begin{aligned}\nabla^2 f(x+h) &= \nabla(\nabla f(x+h) = \nabla(f(x+h) - f(x)) \\ &= \nabla f(x+h) - \nabla f(x)\end{aligned}$$

$$\nabla^2 f(x+2h) = \nabla f(x+2h) - \nabla f(x+h)$$

Third differences:

$$\nabla^3 f(x+h) = \nabla^2 f(x+h) - \nabla^2 f(x)$$

$$\nabla^3 f(x+2h) = \nabla^2 f(x+2h) - \nabla^2 f(x+h)$$

Here we note that,

$$\nabla f(x+h) = f(x+h) - f(x) = \Delta f(x)$$

$$\begin{aligned}\nabla f(x+2h) &= f(x+2h) - f(x+h) \\ &= \Delta f(x+h)\end{aligned}$$

$$\begin{aligned}\nabla^2 f(x+2h) &= \nabla f(x+2h) - \nabla f(x+h) \\ &= \Delta f(x+h) - \Delta f(x) \\ &= \Delta^2 f(x)\end{aligned}$$

In general, $\nabla^n f(x+nh) = \Delta^n f(x)$

Shifting operator (E):

Let $y = f(x)$ be a given function of x and $x_0, x_0+h, x_0+2h, x_0+3h, \ldots, x_0+nh$ be the consecutive values of x. Then the operator E is defined as

$$\mathrm{E}\left[f(x_0)\right] = f(x_0+h)$$

E is called the shifting operator . It is also called the displacement operator.

$$E\left[f\left(x_0+h\right)\right] = f(x_0+2h),$$

$$E[f(x_0+2h)] = f(x_0+3h),\ldots,$$

$$\mathrm{E}[f(x_0+(n-1)h)] = f(x_0+nh)$$

$\mathrm{E}\left[f(x)\right] = f(x+h)$, h is the (equal) interval of spacing

$E^2 f(x)$ means that the operator E is applied twice on $f(x)$

$$\begin{aligned}\text{(i.e)}\quad \mathrm{E}^2 f(x) &= \mathrm{E}\left[\mathrm{E}\, f(x)\right] \\ &= \mathrm{E}[f(x+h) = f(x+2h)\end{aligned}$$

In general ,

$\mathrm{E}^n f(x) = f(x+nh)$ and
$\mathrm{E}^{-n} f(x) = f(x-nh)$

Properties of the operator E:

1. $\mathrm{E}\left[f_1(x) + f_2(x) + \ldots.. + f_n(x)\right] = \mathrm{E}f_1(x) + \mathrm{E}f_2(x) \ldots. + \ldots. + \mathrm{E}\left[f_n(x)\right]$
2. $\mathrm{E}\left[c\, f(x)\right] = cE\left[f(x)\right]$ c is a constant

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